HE2003 Econometrics I
Tutorial 5

16/09/2024

Question 1

(a) Explain the notion of unbiasedness and efficiency by using the sample average $\bar{Y}$ as an example.

The notion of unbiasedness and efficiency is widely applied in all kinds of estimation problems. We take the estimation of population mean μ as an example. Suppose that our first candidate for the estimator is the sample average $\bar{Y}$, and the second candidate is defined as $\tilde{Y} = \frac{1}{2}(Y_1 + Y_n)$ (average of the first and last observations).

We would like to compare the two candidates using the following two criteria: unbiasedness and efficiency. First, as discussed in class, the sample average is unbiased: $E[\bar{Y}] = \mu$, and $\tilde{Y}$ is also unbiased:

$$E[\tilde{Y}] = E\left[\frac{1}{2}(Y_1 + Y_n)\right] = \frac{1}{2}E[Y_1 + Y_n] = \frac{1}{2}(E[Y_1] + E[Y_n]) = \frac{1}{2}(\mu + \mu) = \mu$$

Therefore, both $\bar{Y}$ and $\tilde{Y}$ satisfy the criterion of unbiasedness.

Question 1

Then, we compare the two candidates by using efficiency. As discussed previously, $Var[\bar{Y}] = \sigma^2/n$. On the other hand, we note that:

$$Var[\tilde{Y}] = Var\left[\frac{1}{2}(Y_1 + Y_n)\right] = \frac{1}{4}Var[Y_1 + Y_n] = \frac{1}{4}(Var[Y_1\,] + Var[Y_n]) = \frac{1}{4}(\sigma^2 + \sigma^2) = \frac{1}{2}\sigma^2,$$

where σ^2 is the population variance. Therefore, $\bar{Y}$ is more efficient (has a smaller variance) than $\tilde{Y}$ as long as $n > 2$.

In sum, the sample average is considered a good estimator according to this comparison because both are unbiased but in general the sample average will be more efficient.

Note: for a constant a, we have $Var[aX] = a^2Var[X]$.

$Var[X + Y] = Var[X] + Var[Y]\ + 2Cov[X, Y] = Var[X] + Var[Y]$ if X and Y are independent

Question 1

(b) Explain the Gauss-Markov Theorem for the OLS estimators.

Under the 3 OLS assumptions:

1. The conditional distribution of u_i given X_i has a mean of zero: $E[u_i|X_i] = 0$
2. $(X_i, Y_i), \quad i = 1, \dots, n$ are independently and identically distributed ($i.i.d.$)
3. Large outliers are unlikely (and some other technical assumptions)

The OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are the **best linear unbiased estimators (BLUE)** of β_0 and β_1, respectively.

The Gauss-Markov Theorem states that among all the linear unbiased estimators of β_0 and β_1, the OLS estimator has the **smallest variance (the most efficient).** That is, for any alternative estimators $\tilde{\beta}_0$ and $\tilde{\beta}_1$ that are linear and unbiased, we have $Var[\tilde{\beta}_0] \geq Var[\hat{\beta}_0]$ and $Var[\tilde{\beta}_1] \geq Var[\hat{\beta}_1]$.

Therefore, when the OLS assumptions hold, we need not look for alternative linear unbiased estimator: None will be better than OLS.

Question 2

Suppose that the population mean and variance of employees' working hours per week in Singapore (SG) are μ_1 and σ_1^2, respectively, and N_1 i.i.d. sample observations were drawn from this population; the population mean and variance of employees' working hours per week in Hong Kong (HK) are μ_2 and σ_2^2, respectively, and N_2 i.i.d. sample observations were drawn from this population.

The SG and HK random samples are totally independent. We are interested in $\mu_1 - \mu_2$, the difference between the two population means, and use $\hat{d} = \bar{Y}_1 - \bar{Y}_2$ as its estimator, where $\bar{Y}_1$ is the sample average of the N_1 SG observations and $\bar{Y}_2$ is the sample average of the N_2 HK observations.

*The formulas of means, variances, and covariances of sums of random variables can be found in Key Concept 2.3 (p.74) of "Introduction to Econometrics".

(a) Compute $E[\bar{Y}_1],\ E[\bar{Y}_2],\ Var[\bar{Y}_1],\ and\ Var[\bar{Y}_2]$.

(b) Compute $E[\hat{d}],\ Cov[\bar{Y}_1, \bar{Y}_2],\ and\ Var[\hat{d}]$.

Question 2

$$E[\bar{Y}_1] = E\left[\frac{1}{N_1}\sum_{i=1}^{N_1} Y_{1i}\right] = \frac{1}{N_1}E\left[Y_{11} + Y_{12} + \cdots + Y_{1N_1}\right] = \frac{1}{N_1}\left(E[Y_{11}] + E[Y_{12}] + \cdots + E\left[Y_{1N_1}\right]\right)$$

$$E[\bar{Y}_1] = \frac{1}{N_1}\sum_{i=1}^{N_1} E[Y_{1i}] = \frac{1}{N_1}\sum_{i=1}^{N_1} \mu_1 = \frac{1}{N_1} \times N_1\mu_1 = \mu_1$$

Observations are independent from each other and identical distributed $(i.i.d)$

$$Var[\bar{Y}_1] = Var\left[\frac{1}{N_1}\sum_{i=1}^{N_1} Y_{1i}\right] = \frac{1}{N_1^2}Var\left[Y_{11} + Y_{12} + \cdots + Y_{1N_1}\right] = \frac{1}{N_1^2}\left(Var[Y_{11}] + Var[Y_{12}] + \cdots + Var\left[Y_{1N_1}\right]\right)$$

$$Var[\bar{Y}_1] = \frac{1}{N_1^2}\sum_{i=1}^{N_1} Var[Y_{1i}] = \frac{1}{N_1^2}\sum_{i=1}^{N_1} \sigma_1^2 = \frac{1}{N_1^2} \times N_1\sigma_1^2 = \frac{\sigma_1^2}{N_1}$$

Similarly, we can prove that $E[\bar{Y}_2] = \mu_2$ and $Var[\bar{Y}_2] = \sigma_2^2/N_2$.

Question 2

$$E[\hat{d}] = E[\bar{Y}_1 - \bar{Y}_2] = \mathrm{E}[\bar{\mathrm{Y}}_1] - \mathrm{E}[\bar{\mathrm{Y}}_2] = \mu_1 - \mu_2$$

This implies that $\hat{d}$ is an unbiased estimator for the parameter of interest.

Using the formula $Var[X - Y] = Var[X] + Var[Y] - 2Cov[X, Y]$, we have:

$$Var[\hat{d}] = Var[\bar{Y}_1 - \bar{Y}_2] = Var[\bar{\mathrm{Y}}_1] + Var[\bar{\mathrm{Y}}_2] - 2Cov[\bar{Y}_1, \bar{Y}_2] = Var[\bar{\mathrm{Y}}_1] + Var[\bar{\mathrm{Y}}_2] = \frac{\sigma_1^2}{N_1} + \frac{\sigma_2^2}{N_2}$$

$= 0$ because SG and HK samples are independent from each other.

Question 3

In this question, we show the reason why R^2 cannot be larger than 1.

(a) Show that $\sum_{i=1}^{n} \hat{u}_i X_i = 0$ by using the FOC of the OLS estimators.

(b) Show that $\sum_{i=1}^{n} \hat{u}_i \hat{Y}_i = 0$ [hint: $\sum_{i=1}^{n} \hat{u}_i \hat{Y}_i = \sum_{i=1}^{n} \hat{u}_i \left(\hat{\beta}_0 + \hat{\beta}_1 X_i\right)$.]

(c) Show that $TSS = ESS + SSR$, where TSS and ESS are defined in the lecture slides, while $SSR = \sum_{i=1}^{n} \hat{u}_i^2$

(d) Discuss why the OLS procedure is designed to have $\sum_{i=1}^{n} \hat{u}_i = 0$ and $\sum_{i=1}^{n} \hat{u}_i X_i = 0$.[Hint: There is something in the population that the OLS procedure would like to mimic by using $\sum_{i=1}^{n} \hat{u}_i = 0$ and $\sum_{i=1}^{n} \hat{u}_i X_i = 0$. Also, for two random variables X and Y , the covariance between X and Y can be written as:
$Cov(X, Y) = E\left[(X - \mu_x)\left(Y - \mu_y\right)\right] = E\left[XY - X\mu_y - \mu_x Y + \mu_x \mu_y\right]$
$= E[XY] - E[X]\mu_y - \mu_x E[y] + \mu_x \mu_y = E[XY] - \mu_x \mu_y - \mu_x \mu_y + \mu_x \mu_y = E[XY] - \mu_x \mu_y$
Where $\mu_x = E[X]$ and $\mu_y = E[Y]$.

Question 3

(a) Show that $\sum_{i=1}^{n} \hat{u}_i X_i = 0$

Recall the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are the solution of the minimization problem:

$$\min_{b_0, b_1} \sum_{i=1}^{n} [Y_i - (b_0 + b_1 X_i)]^2$$

We have the second equation of the FOC (taking partial derivatives with respect to b_1):

$$\sum_{i=1}^{n} \left[Y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 X_i\right)\right] X_i = 0$$

$$\Rightarrow \sum_{i=1}^{n} \hat{u}_i X_i = 0$$

Question 3

(b) Show that $\sum_{i=1}^{n} \hat{u}_i \hat{Y}_i = 0$

We have:

$$\sum_{i=1}^{n} \hat{u}_i \hat{Y}_i = \sum_{i=1}^{n} \hat{u}_i\left(\hat{\beta}_0 + \hat{\beta}_1 X_i\right) = \sum_{i=1}^{n} \hat{\beta}_0 \hat{u}_i + \sum_{i=1}^{n} \hat{\beta}_1 \hat{u}_i X_i = \hat{\beta}_0 \sum_{i=1}^{n} \hat{u}_i + \hat{\beta}_1 \sum_{i=1}^{n} \hat{u}_i X_i = 0$$

Due to $\sum_{i=1}^{n} \hat{u}_i = 0$ (proof last week) and $\sum_{i=1}^{n} \hat{u}_i X_i = 0$ (proof in part a)

Question 3

(c) Show that $TSS = ESS + SSR$

We have TSS

$$= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \sum_{i=1}^{n}\left[Y_i - \hat{Y}_i + \hat{Y}_i - \bar{Y}\right]^2 = \sum_{i=1}^{n}\left[\hat{u}_i + \left(\hat{Y}_i - \bar{Y}\right)\right]^2 = \sum_{i=1}^{n}\left\{\hat{u}_i^2 + \left(\hat{Y}_i - \bar{Y}\right)^2 + 2\hat{u}_i\left(\hat{Y}_i - \bar{Y}\right)\right\}$$

$$= \sum_{i=1}^{n}\hat{u}_i{}^2 + \sum_{i=1}^{n}\left(\hat{Y}_i - \bar{Y}\right)^2 + \sum_{i=1}^{n}2\hat{u}_i\left(\hat{Y}_i - \bar{Y}\right) = \sum_{i=1}^{n}\hat{u}_i{}^2 + \sum_{i=1}^{n}\left(\hat{Y}_i - \bar{Y}\right)^2 + 2\sum_{i=1}^{n}\hat{u}_i\hat{Y}_i + 2\sum_{i=1}^{n}\hat{u}_i\bar{Y}$$

Both $= 0$

$$= \sum_{i=1}^{n}\hat{u}_i{}^2 + \sum_{i=1}^{n}\left(\hat{Y}_i - \bar{Y}\right)^2 = SSR + ESS$$

Note: this result implies $R^2 = \dfrac{ESS}{TSS} = \dfrac{ESS}{ESS + SSR}$

Question 3

(d) Discuss why the OLS procedure is designed to have $\sum_{i=1}^{n} \hat{u}_i = 0$ and $\sum_{i=1}^{n} \hat{u}_i X_i = 0$.

The first equation of the FOC of the OLS estimators $\sum_{i=1}^{n} \hat{u}_i = 0$ is equivalent to

$$\frac{1}{n}\sum_{i=1}^{n} \hat{u}_i = 0$$

That is the sample average of the OLS residuals is equal to zero. This is designed to mimic:

$$E[u_i] = 0$$

Which implies the expected value (population mean) of the error terms is equal to zero: the sample average mimics the population mean, and the residuals mimic the error terms.

Question 3

The second equation of the FOC of the OLS estimators $\sum_{i=1}^{n} \hat{u}_i X_i = 0$ is equivalent to

$$\frac{1}{n}\sum_{i=1}^{n} \hat{u}_i X_i = 0$$

That is the sample average of the $\hat{u}_i X_i$ is equal to zero. This is designed to mimic:

$$E[u_i X_i] = 0$$

Which implies the expected value of $u_i X_i$ is equal to zero. Hence, we have:

$$Cov[u_i, X_i] = E[u_i X_i] - E[u_i]E[X_i] = 0$$

Equivalently, $corr[u_i, X_i] = 0$, the error term u_i is uncorrelated with the regressor X_i.

This is important because if u_i and X_i are correlated with each other, then OLS Assumption 1 cannot hold and the OLS estimators will be biased.

Question 4

This question follows the example of law school rankings discussed in the class. The dataset *lawsch_ranking.dta* contains the average starting salaries of students in each US law school, ranking of these law schools (No.1-142), average college GPA of the students in each school, etc.

Now we consider the following model:

$$salary_i = \beta_0 + \beta_1 top10_i + \beta_2 r11\_25_i + \beta_3 r26\_50_i + \beta_4 GPA_i + u_i$$

where $i = 1, \dots, n$. Define the dummy variables $top10$, $r11\_25$, $r26\_50$ to be equal to 1 when the ranking of the i-th school falls into the appropriate range, and 0 otherwise.

(a) Write down precisely the definition of each dummy variable.

(b) Write down the linear regression model for the schools ranked Top 10, 11-25, 26-50, and 51-142, respectively

(c) Explain the meaning of the parameters $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$ by using a graph.

Question 4

(a) Write down precisely the definition of each dummy variable.

$top10_i$ is equal to **one** if the i-th observation is among the Top 10 schools, and is equal to **zero** otherwise.

$r11\_25_i$ is equal to **one** if the i-th observation is among the 11th-25th schools, and is equal to **zero** otherwise.

$r26\_50_i$ equal to **one** if the i-th observation is among the 26th-50th schools, and is equal to **zero** otherwise.

There are totally 4 categories, so we just need 3 dummy variables.

Question 4

(b) Write down the linear regression model for the schools ranked Top 10, 11-25, 26-50, and 51-142

For Top 10 schools: $salary_i = (\beta_0 + \beta_1) + \beta_4 GPA_i + u_i$,

For 11th-25th schools: $salary_i = (\beta_0 + \beta_2) + \beta_4 GPA_i + u_i$,

For 26th-50th schools: $salary_i = (\beta_0 + \beta_3) + \beta_4 GPA_i + u_i$,

For schools ranked below 50th: $salary_i = \beta_0 + \beta_4 GPA_i + u_i$.

Notice that for the case with Top10, 11th-25th, or 26th-50th schools, only one dummy variable will be switched on (equal to one), while the others will be switched off (equal to zero). For the case with schools ranked below 50th (based group), all the three dummy variables will be switched off.

Question 4

Explain the meaning of the parameters $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$ by using a graph.